

WIKIPEDIA

SAKO (programming language)

From Wikipedia, the free encyclopedia

SAKO (PL: *System Automatycznego Kodowania*^[3] - EN: An Automatic Coding System^[4]) is a Polish language-based programming language written between 1959 and 1960 by a team from the ZAM division of the Polish Academy of Sciences.^[1] Originally developed for the XYZ and ZAM-2 computers, it was also ported over to the ZAM-21, ZAM-41 and the Mińsk-22.^[5]

General features of the SAKO language:

- commands similar to sentences used in natural language
- shortening the time required to learn the principles of programming
- ease of use, which reduced coding time
- transparent program code, lowering the probability of making a mistake

It had a static address allocation. It was possible to insert code in SAS macro assembler. The compilation proceeded in two stages:

1. From SAKO to simplified SAS macro assembler (SAS-W).
2. From SAS-W to machine language.

The most characteristic feature of SAKO are Polish commands, e.g. CZYTAJ, SKOCZ DO. It was designed primarily for programming numerical calculations.

"Hello, world" example

```
TEKST:
HELLO, WORLD
LINIA
STOP NASTEPNY
KONIEC
```

References

- Prace Zakładu Aparatów Matematycznych PAN, "System Automatycznego Kodowania SAKO. Cz. 1, Opis języka", PAN – Warszawa 1961
- Leon Łukaszewicz, Antoni Mazurkiewicz "System automatycznego kodowania SAKO" Zakład Narodowy im. Ossolińskich, Polish Academy of Sciences Publishing House, Wrocław-Warszawa-Kraków 1966
- "COMPUTERS AND CENTERS, OVERSEAS: 6. Instytut Maszyn

	SAKO
Paradigm	Multi-paradigm: structured, imperative (procedural), array
Designed by	Leon Łukaszewicz, Antoni Mazurkiewicz, Jan Borowiec, Jowita Koncewicz, Maria Łącka, Stefan Sawicki, Jerzy Swianiewicz, Piotr Szorc, Alfred Szurman and Andrzej Wiśniewski ^[1]
Developer	Polish Academy of Sciences
First appeared	1960
Influenced by	Fortran ^[2]

Matematycznych, ZAM 2, Warsaw, Poland" (<https://web.archive.org/web/20180603153655/http://www.dtic.mil/docs/citations/AD0694639>). *Digital Computer Newsletter*. **16** (1): 22–23. 1964. Archived from the original (<http://www.dtic.mil/docs/citations/AD0694639>) on June 3, 2018.

- Antoni Mazurkiewicz "Arithmetic Formulae and the Use of Subroutines in SAKO" (<https://www.sciencedirect.com/science/article/abs/pii/S0066413861800098?via%3Dihub>) *Annual Review in Automatic Programming*, Vol. 2
- Leon Łukaszewicz "SAKO — An Automatic Coding System" (<https://www.sciencedirect.com/science/article/abs/pii/S0066413861800086?via%3Dihub>) *Annual Review in Automatic Programming*, Vol. 2
- Władysław Turski "Some Results of Research on Automatic Programming in Eastern Europe" *Advances in Computers*, Vol. 5
- "Algorytmy" Vol. 1 No. 1 (<https://bcpw.bg.pw.edu.pl/dlibra/publication/5456/edition/5308>), Institute of Mathematical Machines, Polish Academy of Sciences, Warsaw
- CIA-RDP80T00246A011500230001-8

Citations

1. Łukaszewicz 1966, p. 6
2. Report of a Visit to Discuss Common Programming Languages in Czechoslovakia and Poland, 1963, John A. Gosden (Editor), Roger E. Gay, John L. Jones, Jack N. Merner, Christopher J. Shaw
3. Mazurkiewicz, Antoni (1989). "Jak się programowało XYZ" (<https://dlibra.bg.pw.edu.pl/dlibra/publication/7196/edition/6758>). *Informatyka*. 8-12/1989: 10–12.
4. Łukaszewicz, Leon (1961-01-01). "SAKO—An automatic coding system" (<https://www.sciencedirect.com/science/article/pii/S0066413861800086>). *Annual Review in Automatic Programming*. **2**: 161–176. doi:10.1016/S0066-4138(61)80008-6 (<https://doi.org/10.1016%2FS0066-4138%2861%2980008-6>). ISSN 0066-4138 (<https://search.worldcat.org/issn/0066-4138>).
5. Szmelter, Jan; Zub, Henryk. *Programowanie SAKO na Mińsk-22* (in Polish). Main Library of the Military University of Technology.

Retrieved from "https://en.wikipedia.org/w/index.php?title=SAKO_(programming_language)&oldid=1291206737"

This page was last edited on 19 May 2025, at 19:10 (UTC).

Text is available under the Creative Commons Attribution-ShareAlike 4.0 License; additional terms may apply. By using this site, you agree to the Terms of Use and Privacy Policy. Wikipedia® is a registered trademark of the Wikimedia Foundation, Inc., a non-profit organization.